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GEN AI ARCHITECT JOURNEY · PHASE-003

Playwright Automation

Browser RPA - driving the page, not the screen

Track	Playwright - browser automation
Contents	Two fundamentals scripts and two complete projects
Level	Intermediate
Needs	Python 3.10+ · Playwright · Chromium
Folder	Phase-003-Robotic-Process-Automation-(RPA)/Playwright Automation

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1. Why Playwright is different

Playwright locates page elements through the browser's document structure, not through screen coordinates.

It can wait for navigation and dynamic content, and nothing depends on where a window sits on screen. A resized window, a different resolution, or a second monitor changes nothing - which is exactly what makes it more reliable than desktop automation for anything that lives in a browser.

2. The fundamentals scripts

Script	What it demonstrates
Playwright_basic.py	Opens Google in headless Chromium, saves google.png, closes the browser
Playwright_Functionality.py	A commented reference of common Playwright capabilities - it does not execute automation

3. Setup

```
POWERSHELL
cd "Playwright Automation"
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install playwright
python -m playwright install chromium

python Playwright_basic.py
```

The browser is headless, so no window appears. A successful run creates **google.png** in the current working directory.

4. Concepts covered

- Starting and closing browsers and contexts
- Headed and headless execution
- Page navigation and URL waits
- Locating page elements using stable attributes
- Filling inputs and clicking controls
- Extracting text and metadata
- Taking full-page screenshots
- Reusing a persistent browser profile
- Handling timeouts, dynamic content, and selector fallbacks
- Producing JSON and Excel automation reports



5. The two complete projects

Project	What it automates
Playwright Automation Project	SauceDemo product extraction -> screenshots -> Excel report
Playwright Assignment	Consent-based WhatsApp messaging, reply monitoring, and reports

Install per project, not globally

The two subprojects have different dependencies and different operational risks. Follow the README in the project you actually want to run rather than installing everything into one environment.

6. Troubleshooting

Symptom	Fix
Chromium is missing	Run python -m playwright install chromium in the active environment
Browser installation is blocked	Ensure the terminal may download and execute Playwright browsers
A page element cannot be found	The site's HTML may have changed - inspect and update the locator
You need to watch it run	Set \$env:PWDEBUG='1' before the command to open Playwright Inspector

Next

Continue to the SauceDemo Product Automation, the first complete browser project in this track.

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